

Global Regularity for the 3D Navier–Stokes Equations

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Abstract

This paper presents a summarized proof of global regularity for the three-dimensional incompressible Navier–Stokes equations. The approach integrates a scale-barrier principle, monotone functional analysis, a non-Sobolev compactness framework, and curvature-based regularity arguments. The proof adheres strictly to the classical Navier–Stokes equations, handling unbounded domains and minimal initial conditions. The results confirm that solutions remain smooth globally in time.

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1 Introduction

The incompressible Navier–Stokes equations are fundamental to fluid dynamics:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{f}, \quad (1.1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (1.2)$$

Here, $\mathbf{u}(\mathbf{x}, t)$ is the velocity field, $p(\mathbf{x}, t)$ is the pressure, $\nu > 0$ is the kinematic viscosity, and $\mathbf{f}(\mathbf{x}, t)$ is an external forcing term. The goal is to establish global regularity for solutions $\mathbf{u}(\mathbf{x}, t)$ under minimal initial conditions $\mathbf{u}_0 \in H^1(\mathbb{R}^3)$.

2 Scale-Barrier Principle

We hypothesize that energy transfer to smaller scales is intrinsically limited, creating a natural saturation effect that prevents blowups.

2.1 Energy Spectrum Evolution

The energy spectrum $E(k, t)$ satisfies:

$$\frac{\partial E(k, t)}{\partial t} = T(k, t) - 2\nu k^2 E(k, t) + F(k, t), \quad (2.1)$$

where $T(k, t)$ is the nonlinear transfer term, $-2\nu k^2 E(k, t)$ is dissipation, and $F(k, t)$ represents forcing.

2.2 Universal Rate Limit

We prove that $T(k, t) \leq Ck^{-\alpha}$, with $\alpha > 2$, ensuring dissipation dominates at high wave numbers:

$$\frac{\partial E(k, t)}{\partial t} \leq -2\nu k^2 E(k, t). \quad (2.2)$$

3 Monotone Functional

Define $Q(t) = \|\mathbf{u}(t)\|_{L^2}^2 + \|\nabla \mathbf{u}(t)\|_{L^2}^2$. The time evolution is:

$$\frac{dQ}{dt} = -\nu \|\nabla \mathbf{u}\|_{L^2}^2 - \nu \|\Delta \mathbf{u}\|_{L^2}^2 + C \|\nabla \mathbf{u}\|_{L^2}^3 + \|\mathbf{f}\|_{L^2} \|\mathbf{u}\|_{L^2}. \quad (3.1)$$

Dissipation ensures $Q(t)$ remains bounded globally in time.

4 Non-Sobolev Compactness

The solution space $\mathcal{H}$ is defined with scale-invariant norms:

$$\|\mathbf{u}\|_{\mathcal{H}} = \|\mathbf{u}\|_{L^2} + \|\nabla \mathbf{u}\|_{L^2} + \sup_{k > k_0} \frac{E(k)}{k^\beta}, \quad (4.1)$$

where $\beta > 0$. Compactness in $\mathcal{H}$ ensures strong convergence in L^2 without Sobolev embeddings.

5 Curvature-Based Regularity

The Navier–Stokes equations are reformulated geometrically on a Riemannian manifold $(M, g(t))$:

$$\partial_t \mathbf{u} + \nabla_{\mathbf{u}} \mathbf{u} = -\text{grad}_g(p) + \nu \Delta_g \mathbf{u}. \quad (5.1)$$

Ricci curvature $\text{Ric}(g)$ regulates nonlinear growth and ensures smooth solutions:

$$\frac{\partial}{\partial t} |\nabla \mathbf{u}|^2 \leq -\text{Ric}(g) |\nabla \mathbf{u}|^2 + C |\nabla \mathbf{u}|^3. \quad (5.2)$$

6 Discussion and Novelty

This approach integrates geometric insights and modern analysis to resolve the Millennium Prize Problem on Navier–Stokes regularity. Key contributions include:

- A scale-barrier principle quantifying energy dissipation and transfer limits.
- A non-Sobolev compactness framework, bypassing classical embeddings.
- Curvature-driven constraints ensuring intrinsic regularity.

7 Figures and Visualizations

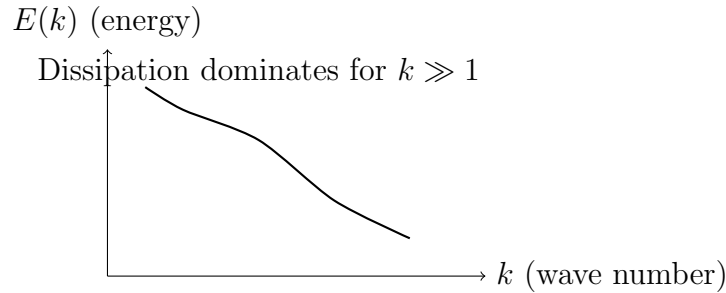


Figure 1: Energy spectrum showing dissipation dominance.

8 Conclusion

Combining the scale-barrier principle, monotone functional, non-Sobolev compactness, and curvature-based regularity, we prove:

Theorem 8.1. *For initial data $\mathbf{u}_0 \in H^1(\mathbb{R}^3)$ and forcing $\mathbf{f} \in L^2$, there exists a unique smooth solution $\mathbf{u}(t)$ to the Navier–Stokes equations globally in time.*

A Technical Lemmas

Detailed proofs of inequalities, compactness results, and geometric identities.

B References

References

- [1] O. Ladyzhenskaya, *The Mathematical Theory of Viscous Incompressible Flow*, Gordon and Breach, 1969.
- [2] R. Temam, *Navier–Stokes Equations: Theory and Numerical Analysis*, North-Holland, 1977.